

Interpretation of Verification Results for Existential and Hybrid Properties

Figure 1 shows how MELA interprets satisfaction and violation results returned by NUSMV for existential properties. For existential properties, MELA uses an interpretation dual to the one used for universal properties. In CTL, an existential requirement can be written, for example, as $EF(\varphi)$, which states that there exists at least one path along which a state satisfying φ is eventually reached. If NUSMV reports that the learned Moore machine violates such a requirement, then no execution in the learned machine satisfies the required behaviour. Since the behaviours represented by the PTA are contained in the learned machine, the PTA cannot provide an observed execution satisfying the property either, and MELA reports the requirement as violated. If the learned Moore machine satisfies the requirement, MELA checks whether the same existential requirement is also satisfied by the PTA. To do so, MELA verifies the same CTL property on the PTA encoded in NUSMV. If the PTA also satisfies the property, then the required existential behaviour is present in the observed traces, and MELA reports the requirement as satisfied. If the PTA violates the property, then the satisfying behaviour appears only in the learned Moore machine and is attributed to state merging; in this case, MELA reports the requirement as violated.

For LTL, the existence of an execution satisfying a formula ψ is checked through the universal complement. Specifically, MELA verifies $\neg\psi$ as an LTL specification. If NUSMV violates $\neg\psi$, the returned counterexample is an execution satisfying ψ , and the existential requirement is satisfied on the learned Moore machine. MELA then performs the same check on the PTA. If $\neg\psi$ is also violated on the PTA, then the existential witness is supported by the observed traces, and MELA reports the requirement as satisfied. If $\neg\psi$ is satisfied on the PTA, then the existential witness exists only in the learned Moore machine and is attributed to state merging.

For existential implication properties, MELA applies the same vacuity principle used for universal implication properties: satisfaction is considered vacuous when the condition that should trigger the required behaviour is never reached in the machine.

Hybrid properties contain both universal and existential path quantifiers. For such properties, MELA interprets the result recursively, following the structure of the formula. Universal subformulas are handled using the universal interpretation, while existential subformulas are handled using the dual exis-

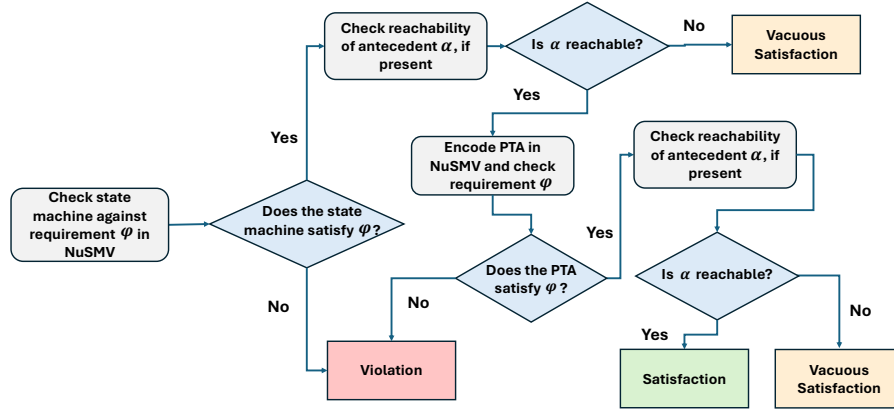


Figure 1: MELA's process in Step 6 for interpreting the verification results for existential properties.

tential interpretation. In each case, the PTA is used to distinguish behaviours supported by observed traces from behaviours introduced by state merging.